Capstone Analysis

Designing for Course Readiness: Instructional Design Decisions for T127 Microlearning

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Introduction

This course has made me realize that, learning design is all about making choices. The analysis that I present here is my attempt to be honest about the choices I made while designing two modules of the pre-course microlearning unit for incoming T127 students: a module on Higher-Ed Learning design in the Teaching Learning Lab (TLL) context and one on Portfolio planning.

The microlearning unit was designed as a six-frame learning journey for incoming T127 students: a welcome and course roadmap, an introduction to higher-ed learning design and the TLL context, practicum readiness and ways of working, project direction, portfolio planning, and a closing reflection with feedback. The full unit was a team effort. My individual contribution covers two of those frames: Frame 2, higher-ed learning design and the TLL context, and Frame 5, portfolio planning. The design decisions I write about here are entirely my modules. What I kept, what I removed, what I chose to build in HTML rather than H5P, and why I put video in one place but not another. Many of them happened in the middle of building, testing, and testing things. What the readings gave me was a way to name what had happened after the fact and increasingly, to anticipate decisions before I made them.

Target audience influence

The audience for this unit is incoming T127 students: a graduate-level cohort at HGSE who are arriving with varied backgrounds and no prior experience with the TLL. They are encountering this Microlearning material before the course starts, which means they have no frame of reference for the course, TLL's practices, or what project-based learning at this level actually feels like.

Audience profile was the core of my design choices from the beginning. It is based on the course's principle is that: "if you do not design for someone, you are designing for no one". When I originally built Activity 1, the myth-and-reality flip cards about the TLL, I included a confidence rating prompt before each card. The idea was to encourage metacognition: students would rate their certainty before seeing the answer. But when I tested it, I realized that these students genuinely know nothing about the TLL before the course begins. Asking them to rate their confidence about something they have no frame of reference for does not create metacognition. It creates an extra step and hence removing the confidence rating was a direct response to who the learner was at that moment.

I also reduced the number of flip cards from five to three as a novice learner needs structured cues, not an exhaustive list (Persky and Robinson's, 2017). I kept only the myths most likely to genuinely challenge an assumption an incoming student would hold.

We conducted a needs analysis in the form of an online survey, though only five students responded. Even with that small sample, a few patterns were useful. Four of the five described themselves as "somewhat familiar" with practicum courses, meaning that they were not starting from zero but also not confident about what this course specifically involved. The most common attribute across all five responses was capstone ambiguity that includes choosing a project and navigating the development of the project. The students also needed clarity on the workload and applying learning theory to real design work. This indicated that the students needed the pre-course unit to reduce uncertainty.

The responses also reinforced keeping the experience light by mentioning the duration to be between 30 minutes and 2 hours of pre-course work felt reasonable. This helped in limiting the scope of each module rather than trying to cover everything. In my project, even limited data helped sharpen what serving them actually meant. Even a small survey confirmed that students arrive with uneven familiarity with project-based learning, which supported keeping the early sections of the module orientation-focused rather than front-loading expectations.

Learning Outcomes

As this Microlearning is a pre-course experience, the outcomes are oriented toward readiness rather than mastery. The unit was structured as a six-frame learning journey, and my two modules were placed at specific points within it. Out of the six modules, the higher-ed learning design in TLL context module, was placed early in the sequence, where the learner is still orienting. In the fifth module, portfolio planning, came later as an action item. These modules in the Microlearning project defines the scope of its outcomes. The first outcome needed to be about reframing assumptions, not building knowledge. The second needed to prompt a concrete behavior, not explain a concept.

For my two sections, the outcomes I designed toward were:

- Students will be able to distinguish common misconceptions about the TLL from the reality of how the practicum course operates.
- Students will feel oriented to the purpose of the T127 portfolio before the course begins, and will understand what kind of work they will be expected to document.

Wiggins and McTighe's (2005) backward design framework ask the designer to begin with the desired outcome, then determine acceptable evidence, and only then plan the learning experience. Working backward from these two outcomes made it easier to resist adding more content. The myth-busting activity serves the first outcome. The video in the portfolio section serves the second. Everything I removed, including the confidence ratings and two of the five flip cards, was because it did not add value to these outcomes in a clear way.

Instructional Process and Learning Theory

I wanted to follow the constructivist instructional approach. The idea was to make the learners to actively construct the knowledge rather than passively receive the information. Ertmer and Newby (1993) introduced earlier in the course gave me a language for this distinction. The flip card format was chosen because it creates a small moment of active engagement. The student encounters a claim, processes it, and then flips to test or revise their understanding. Even after removing the confidence rating, the flip card still asks the learner to do something before receiving information rather than simply reading a list of facts.

At the same time, a pre-course microlearning unit is not the right for open-ended inquiry as the incoming students might not yet have enough domain knowledge for that to be productive. The design needed to have constructivist principles that are appropriate for novices in the form of structured and scaffolded interactions. Managing cognitive load as mentioned by Young, 2022 where the difference between intrinsic load (the complexity of the content itself), extraneous load (complexity added by poor design), and germane load (the productive effort that builds schema) was also considered as a part of the design choice. Reducing the flip cards from five to three was as much a cognitive load decision from the audience perspective. Each extra card was adding unnecessary mental effort for a learner who was still trying to form a basic picture of what the TLL course is.

The portfolio module presented a different challenge. I was not trying to teach a concept. I was trying to motivate a behavior: to help students feel oriented to the portfolio before the course starts. For that purpose, I chose video. Bates (2022) mentions that different media serve different pedagogical purposes. A conversational, motivational message is better delivered through video than structured text as it would have felt like documentation. The final version uses an AI-generated voice because the instructor-recorded version was not available. I recognize that limitation. A real instructor voice would serve the motivational intent better, and that remains a useful next step for the unit.

Media Choices and Platform

The platform for the unit is SWIFT, Harvard's Learning Experience Platform (LXP). This was a team-level decision. The LXP's HTML teaching element supports custom HTML, CSS, and JavaScript, but has constraints around page structure and how styles interact when multiple activities share the same page.

I built my sections in HTML, CSS, and JavaScript rather than H5P. The flip card activity had a specific visual identity and interaction logic that I was not confident H5P templates could accommodate without significant custom work. Building in HTML gave me precise control over how the activities looked and functioned. The format choices were also supported by the survey data. Short videos and quick cards or slides were the two most preferred formats among respondents. This did not determine the design, but it gave some grounding to make design decisions.

Building on HTML also gave the option to maintain visual consistency across my pages, including the color palette and card styles. But the tradeoff is real as H5P content is more portable and accessible by default, and easier to hand off to another designer. My HTML approach requires the designer to follow a practical long-term path by resolving the CSS consistency issues across HTML pages rather than rebuilding everything, but that work is still in progress.

One structural problem that I encountered during development was style collisions. When multiple activities shared the same LXP page, they used generic class names and the browser could not distinguish between them. I solved this by prefixing every class and ID with an activity-specific label. This kept structure, style, and behavior conceptually separate. When something broke, I knew exactly where to look.

Assessment

Microlearning is a pre-course unit so, there is no formal graded assessment. The flip card activity functions as a form of embedded assessment. When a student flips a card and discovers that a belief they held was a misconception, that is a small check for understanding.

For the portfolio module, the outcome is behavioral: does the student arrive to the first class with a portfolio platform set up and a sense of what belongs in it? That is harder to measure in a pre-course context.

Reflection

Looking back, I notice that my decisions got sharper as the project moved forward. Early on, I was asking what should go in here. By the end, I was asking what does this element do for this learner at this moment. That shift, from content coverage to learner experience, is what made the design work more honest. Every element in a design is a claim about what a learner needs, and that claim has to be grounded in something. The readings gave me a vocabulary for those claims and the project gave me the opportunity to actually make them.

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